

1 Additional Standard Results

Proposition 1. *Let $f : \mathbb{R}_{++} \rightarrow \mathbb{R}_+$ be a continuous function. Consider sequences x^n in $\mathbb{R}_{++}$ and $f^n(x^n)$ in $\mathbb{R}_+$. If $f^n(x^n) \rightarrow f(x)$ as $n \rightarrow \infty$, then $x^n \rightarrow x$ as $n \rightarrow \infty$.*

Proof. Given that f is continuous at x (with $x \in \mathbb{R}_{++}$), for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all y in $\mathbb{R}_{++}$ with $|y - x| < \delta$, we have $|f(y) - f(x)| < \epsilon$.

Given $f^n(x^n) \rightarrow f(x)$, for the above ϵ , there exists an N such that for all $n > N$, $|f^n(x^n) - f(x)| < \epsilon$.

Assume for the sake of contradiction that x^n doesn't converge to x . This implies that there exists a $\delta > 0$ such that for infinitely many terms of the sequence x^n , $|x^n - x| \geq \delta$.

By the continuity of f at x , if $|x^n - x| \geq \delta$ for infinitely many n , then $|f^n(x^n) - f(x)| \geq \epsilon$ for those n , contradicting our assumption that $f^n(x^n) \rightarrow f(x)$.

Therefore, our assumption for contradiction is false, and it follows that $x^n \rightarrow x$ as $n \rightarrow \infty$. $\square$

Fact 1. *Let $g : X \rightarrow \mathbb{R}_+$ be a continuous function, where $X \subseteq \mathbb{R}^n$ is an open convex set. Define the weighted supremum norm $\|\cdot\|_g$ of a real-valued function $f : X \rightarrow \mathbb{R}$ by*

$$\|f\|_g := \sup_{x \in X} \frac{|f(x)|}{g(x)}. \quad (1)$$

If $\lim_{n \rightarrow \infty} \|f_n - f^\|_g = 0$, f_n converges to f^* uniformly on compact sets.*

Proof. Let $\tilde{X}$ be an arbitrary compact subset of X . Since $\tilde{X}$ is compact, there exists a positive lower bound for g on this subset, denoted as

$$\bar{g} = \min_{x \in \tilde{X}} g(x) > 0. \quad (2)$$

Hence, on $\tilde{X}$, if $\lim_{n \rightarrow \infty} \|f_n - f^*\|_g = 0$, then $\lim_{n \rightarrow \infty} \|f_n - f^*\|_\infty = 0$ on $\tilde{X}$, where $\|\cdot\|_\infty$ denotes the supremum norm.

Now, let K be a compact subset of X . Given the continuity of g , there exists a positive maximum value for g on K , denoted as M_K . Then, we have

$$\sup_{x \in K} |f_n(x) - f(x)| \leq M_K \sup_{x \in K} \frac{|f_n(x) - f(x)|}{g(x)} \leq M_K \sup_{x \in X} \frac{|f_n(x) - f(x)|}{g(x)}. \quad (3)$$

Thus, $\lim_{n \rightarrow \infty} \|f_n - f\|_\infty = 0$ implies that f_n converges uniformly to f on the compact set K . It's also worth noting that the convexity and openness of X aren't strictly necessary for this argument. $\square$

Fact 2. *Let $\{f_n\}$ be a sequence of continuous functions defined on a subset of the real line and converging uniformly to a function f on compact sets. If $\{x_n\}$ is a convergent sequence of real numbers with limit x , then $f_n(x_n)$ converges to $f(x)$.*

Proof. Let $\tilde{X}$ be an arbitrary compact subset of X . Since $\tilde{X}$ is compact, there exists a positive lower bound for g on this subset, denoted as

$$\bar{g} = \min_{x \in \tilde{X}} g(x) > 0. \quad (4)$$

Hence, on $\tilde{X}$, if $\lim_{n \rightarrow \infty} \|f_n - f^*\|_g = 0$, then $\lim_{n \rightarrow \infty} \|f_n - f^*\|_\infty = 0$ on $\tilde{X}$, where $\|\cdot\|_\infty$ denotes the supremum norm.

Now, let K be a compact subset of X . Given the continuity of g , there exists a positive maximum value for g on K , denoted as M_K . Then, we have

$$\sup_{x \in K} |f_n(x) - f(x)| \leq M_K \sup_{x \in K} \frac{|f_n(x) - f(x)|}{g(x)} \leq M_K \sup_{x \in X} \frac{|f_n(x) - f(x)|}{g(x)}. \quad (5)$$

Thus, $\lim_{n \rightarrow \infty} \|f_n - f\| = 0$ implies that f_n converges uniformly to f on the compact set K . It's also worth noting that the convexity and openness of X aren't strictly necessary for this argument. $\square$

Claim 1. *If f is convex and $f < 0$ on $(0, \lambda)$, then $\frac{f(s)}{s}$ is increasing on $(0, \lambda)$.*

Proof. Let f be convex on $(0, \lambda)$ and $f < 0$ on $(0, \lambda)$. Let x_1 and x_2 be two points in $(0, \lambda)$. Choose $0 < \alpha < x_1$. Then, any point (in particular, x_1) in (α, x_2) can be written as $x_1 = t\alpha + (1-t)x_2$ for some $0 < t < 1$.

Now, define a new function F on $[\alpha, x_2]$ as:

$$F(x) = f(x) - f(\alpha).$$

Since f is convex, $F(x)$ is also convex on $[\alpha, x_2]$. To see this, observe that:

$$F(t\alpha + (1-t)x_2) = f(t\alpha + (1-t)x_2) - f(\alpha) \leq tf(\alpha) + (1-t)f(x_2) - f(\alpha) = tF(\alpha) + (1-t)F(x_2).$$

Since $F(\alpha) = 0$, the inequality simplifies to $F(x_1) \leq (1-t)F(x_2)$. This implies that $\frac{F(s)}{s}$ is increasing. And thus, if $y_1 \leq y_2$, then:

$$\frac{F(y_1)}{y_1} \leq \frac{F(y_2)}{y_2}.$$

Now, using the definition of $F(x)$, we have:

$$\frac{f(y_1)}{y_1} = \frac{F(y_1)}{y_1} + \frac{f(\alpha)}{y_1}.$$

Similarly, for y_2 :

$$\frac{f(y_2)}{y_2} = \frac{F(y_2)}{y_2} + \frac{f(\alpha)}{y_2}.$$

Since $\frac{F(y_1)}{y_1} \leq \frac{F(y_2)}{y_2}$ and $f(\alpha) < 0$, we conclude that:

$$\frac{f(y_1)}{y_1} \leq \frac{f(y_2)}{y_2}.$$

Thus, $\frac{f(s)}{s}$ is increasing on $(0, \lambda)$, completing the proof. □